

One definition. Every team. *The same answer.*

Governed conversational analytics on Snowflake semantic views, with a test that proves the answers stay identical.

TEAM NAME	TEAM LEADERS	TEAM SIZE	PROBLEM STATEMENT
3M-ONTOLOGIST	Mahantesh · Devansh · Sathyam	3	Teams compute the “same” supply chain metric differently, get different numbers, and cannot prove which is right. The wrong one quietly clears non-compliant suppliers.

1 · PROBLEM BRIEF

Two numbers.
Same 420,000
receipt lines.

“Same question, two answers. The hour gets spent on whose number is right, not on what to do.”

LEGACY

AVERAGE OF MONTHLY AVERAGES

0.882631

Supplier on-time delivery, as most dashboards compute it today.

Δ 0.006807 — enough to change who passes

GOVERNED

EVERY RECEIPT LINE COUNTS ONCE

0.875824

Defined once as a Snowflake semantic view, tested against atomic facts.

Source SEMANTIC.SC_SUPPLIER_LEGACY_DEFECT vs SC_SUPPLIER

1 · PROBLEM BRIEF

A wrong pass makes *no noise*.

BUSINESS PROBLEM

One metric, a different number in every team's tool. The disagreement is found in a review meeting, not in a test.

TARGET USERS

Planning

Procurement

Logistics

Logistics · EU

Ontology steward

PAIN → FIX

~~Hours arguing over whose number is right.~~ Metrics defined once as semantic views and proven identical for every persona, every day.

INDUSTRY

Manufacturing supply chain. ERP, WMS, TMS, IoT and supplier contracts, each with its own idea of “on-time”.

12 / 300

suppliers wrongly cleared by the legacy definition

+22%

bigger “compliant” list than the truth supports

0

suppliers wrongly escalated, so nobody complains

A false fail is disputed within a week. A false pass is never rechecked.

It *tests its own agreement*, every day.

And keeps a deliberately broken metric deployed, so you can watch the test catch it.

	TYPICAL BI DASHBOARD	TYPICAL TEXT-TO-SQL CHAT	SUPPLY CHAIN ONTOLOGY
Where a metric is defined	In each report	In each prompt	Once, in a governed registry
Same number for every team	Hoped for	Not guaranteed	Proven daily, zero spread
Access control	App layer	App layer	Snowflake role and row policy
Numbers in the answer	Not applicable	Can be invented	Invented figures discarded
Contracts and free text	Left out	Left out	Read by AI_EXTRACT, 2,700 of 2,700 fields
Cost of disagreement	Unknown	Unknown	12 suppliers · \$178,832 hidden

Plain English in. *A governed number out.*

The model never writes SQL. It picks registered metrics; anything unregistered cannot be asked.

Follow-ups work. “And by supplier region?” is understood, and every turn is re-checked against the persona’s grants.

Invented figures are discarded. Prose with a number the rows lack is replaced by a template.

Speak it. Whisper runs in the browser. Audio never leaves the device.

1.7 s

93.3%

repeat question, was 16–20 s

on a scored 60-question eval

sc-ontology-app.vercel.app/ask

Period · Trailing 12 months · 2025-09-01 to 2026-08-31 (12 complete months)

What is our supplier on-time delivery?
as Ontology Steward · Trailing 12 months

GOVERNED ANSWER

The question asks for an overall supplier on-time delivery figure, which is the inbound metric measuring receipt lines received on or before the supplier’s promised date.

1 row

executed as SC_ONTOLOGY_STEWARD (secondary roles disabled, row scope ALL REGIONS)

certified · 13/13 quality checks · 15/15 drift-free

 Trailing 12 months

Supplier On-Time Delivery for the trailing 12 months stands at 87.17%. The target for this metric is 90.00%, which is not met. This metric reflects REALIZED scope data from the governed registry.

Supplier On-Time Delivery

drift PASS

87.17%

target 90.00% · below target

Share of purchase-order receipt lines received on or before the date the supplier promised. Weighted by receipt line, so a supplier with many lines influences the result in proportion to its volume.

Show the rows behind this

● Governed Forward Outlook (as_of_scope: PREDICTED)

TARGET_BREACH

Accuracy on held-out data: 99.3%

Interactive simulation ›

Films Breach probability 87.42% 100.0% (UNREACHABLE)	Abrasives Breach probability 86.94% 100.0% (UNREACHABLE)	Adhesives Breach probability 87.24% 99.9% (UNREACHABLE)
Safety Breach probability 87.05% 99.9% (UNREACHABLE)	Filtration Breach probability 87.05% 99.9% (UNREACHABLE)	Tapes Breach probability 87.17% 99.8% (UNREACHABLE)
Medical Breach probability 87.35% 99.7% (UNREACHABLE)	Electronics Breach probability 87.48% 99.0% (UNREACHABLE)	

Target is 5.12 standard deviations of ordinary month-to-month variation BEYOND current capability for Films (mean 0.8742, sd 0.005 over 22 full months). The z-score and verdict are the reliable outputs; the probability is a logistic approximation to the normal CDF and 22 monthly observations cannot support precise tail probabilities.

One metric. Four real roles. *The same number.*

Executed as each Snowflake role, with secondary roles off, so the database enforces access.

Checked against atomic SQL. Two views cannot agree while both being wrong. Runs daily at 06:00 UTC.

Drill-down obeys the same role. Exception rows run as the signed-in persona, so row policy and cost masking apply; an unmapped user gets 403.

A broken control stays live. 0.882631 against the correct 0.875824 proves the test can fail.

15 / 15 4

bindings match, zero spread

personas, same answer

sc-ontology-app.vercel.app/consistency

GOVERNED ANALYTICS WORKSPACE

Cross-Persona Consistency

Compare governed metrics across real Snowflake roles and row scopes.

PERSONAS

4

Real roles, enforced by Snowflake RBAC

SHARED METRICS

15

Served by more than one semantic view

CURRENT AGREEMENT

EXACT

Zero spread across all bindings

DIVERGENCE BEFORE FIX

0.68%

Supplier OTD, same question, two answers

Live cross-persona execution

Pick a governed metric. The app re-executes it once per persona role, through each semantic view that persona can reach, using that role's own grants. Identical values mean the definition — not the team — decides the answer.

GOVERNED METRIC

Supplier On-Time Delivery

Run as all personas

Share of purchase-order receipt lines received on or before the date the supplier promised. Weighted by receipt line, so a supplier with many lines influences the result in proportion to its volume.

Persona access model

Persona	Snowflake role	Focus	Row scope	Semantic views
Ontology Steward	SC_ONTOLOGY_STEWARD	Owns the semantic layer. The only persona that sees every view, the drift control and the consistency evidence.	ALL REGIONS	SC_CONTRACT, SC_DEMAND, SC_FULFILLMENT, SC_INVENTORY, SC_LANDED_COST, SC_MANUFACTURING, SC_ONTOLOGY_360, SC_SUPPLIER, SC_TELEMETRY
Procurement Analyst	SC_PROCUREMENT	Supplier delivery, fill and purchase price variance on the inbound side.	ALL REGIONS	SC_CONTRACT, SC_ONTOLOGY_360, SC_SUPPLIER
Logistics Manager	SC_LOGISTICS	Outbound delivery performance and cost to serve, all regions.	ALL REGIONS	SC_FULFILLMENT, SC_LANDED_COST, SC_ONTOLOGY_360, SC_TELEMETRY
Logistics Manager, EU	SC_LOGISTICS_EU	The same accountability as SC_LOGISTICS, restricted to EU destinations. Same metric definitions, different rows.	SHIP_REGION = EU	SC_FULFILLMENT, SC_LANDED_COST, SC_ONTOLOGY_360, SC_TELEMETRY
Demand & Inventory Planner	SC_PLANNER	Inventory cover, stock value and demand plan accuracy.	ALL REGIONS	SC_DEMAND, SC_FULFILLMENT, SC_INVENTORY, SC_ONTOLOGY_360

Row scope and metric definition are deliberately separate concerns. The EU-scoped logistics role sees fewer rows because of a row access policy,

FEATURE 03 · ONTOLOGY

The diagram *cannot drift* from what is queried.

Read live from INFORMATION_SCHEMA. Entities and relationships come from the deployed view.

Hierarchies are measured. Every rollup is tested as a true functional dependency. Two plausible ones were rejected.

Portable. OWL/SKOS export as Turtle or JSON-LD, with SCOR codes as exactMatch and closeMatch.

12

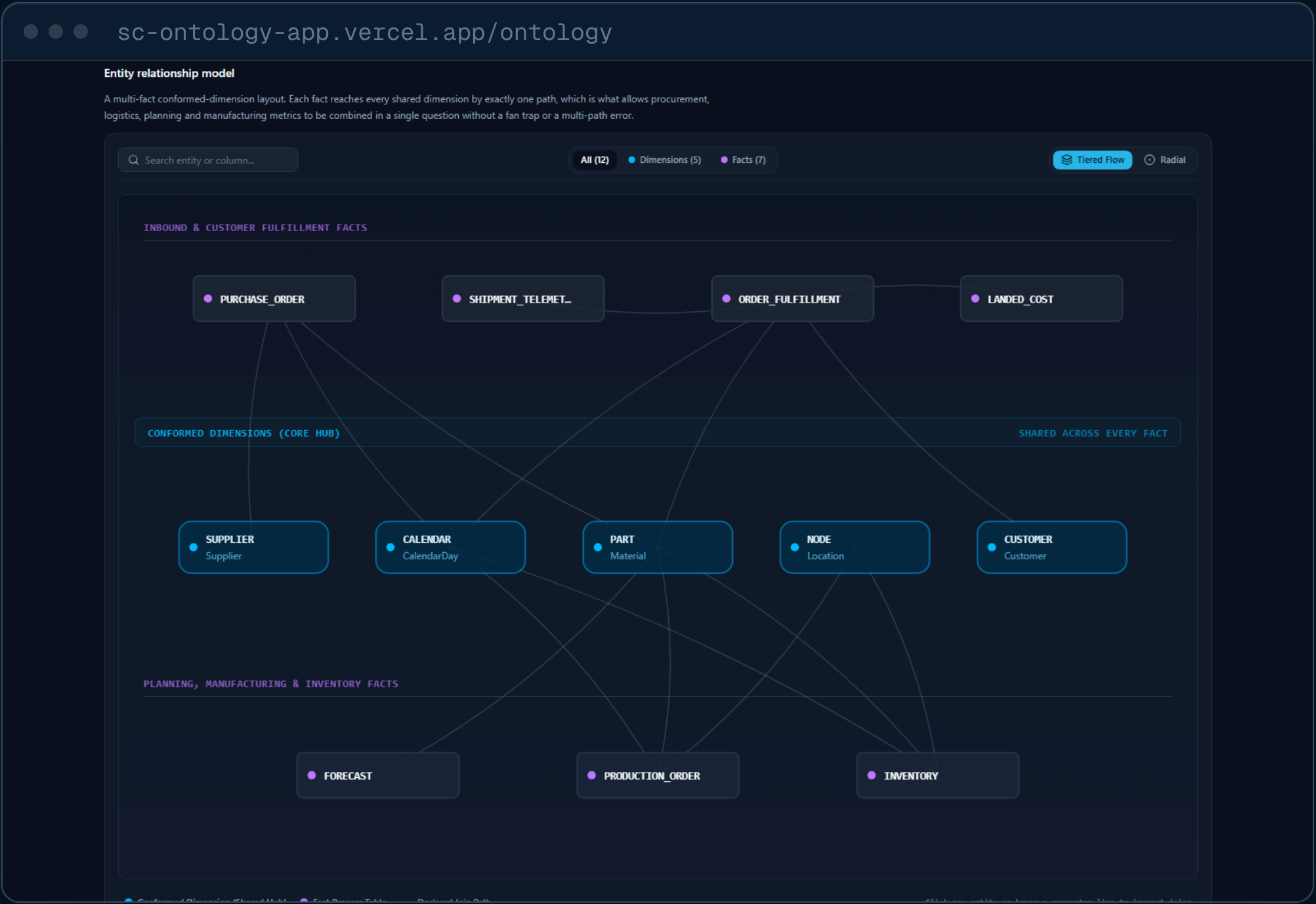
entities

15

relationships

10

rollups proven



sc-ontology-app.vercel.app/impact

Impact

What governed definitions are worth, measured: consistency, accuracy, decisions, money and time, each with its source.

Every figure below is read from a governed object at page load and carries its source. Exactly one input is an assumption: how long a disputed cross-team metric takes to settle by hand. It is labelled, and the one row derived from it says so.

Pillar	Measure	Value	Basis	Source
Consistency	Metric bindings resolving to the canonical value	15 / 15	MEASURED	METRIC_DRIFT_RESULT (run 2026-09-29)
Consistency	Largest spread between personas on the same metric	0	MEASURED	METRIC_DRIFT_RESULT
Accuracy	Conversational eval pass rate (golden questions, run as each persona)	90% (54 / 60)	MEASURED	AGENT_EVAL_RUN (claude-sonnet-4-5)
Accuracy	Invented-metric questions correctly refused	7 / 8	MEASURED	AGENT_EVAL_RUN
Accuracy	Cortex Agent reproduces the canonical definition (agent vs. app vs. registry SQL)	4 / 4 reconciled, 0 diverge (4 differ only by the governed as-of rule)	MEASURED	AGENT_PARITY_RESULT
Accuracy	Contract terms extracted correctly by AI_EXTRACT	100.0% of fields	MEASURED	V_CONTRACT_EXTRACTION_ACCURACY
Decisions	Suppliers the legacy definition wrongly clears	12 of 300 (compliant list inflated 22%)	MEASURED	V_DIVERGENCE_IMPACT
Money	Claimable contract penalties on governed breaches	\$815,222 across 67 suppliers	MEASURED	V_CONTRACT_IMPACT
Money	Penalties hidden by the legacy definition (breached, shown compliant)	\$178,832 across 18 suppliers	MEASURED	V_CONTRACT_IMPACT
Governance	Contracts measuring OTD with the non-governed definition	79 of 300	MEASURED	V_CONTRACT_IMPACT
Time	Mean time to a governed, persona-scoped answer	18.0 s (p95 20.1 s)	MEASURED	AGENT_EVAL_RUN
Time	Manual reconciliation of a disputed cross-team metric (baseline)	4 h	ASSUMPTION	replace with your own figure
Time	Speed-up versus the manual baseline	799x faster	ASSUMPTION	row 11 measured / row 12 assumed

How the scorecard is assembled

SOL PROVENANCE >

FEATURE 04 · IMPACT

Every claim carries *its source*.

MEASURED or ASSUMPTION, labelled on every row, with the object it came from.

Contracts become money. AI_EXTRACT reads 300 free-text contracts and holds them against the governed metric.

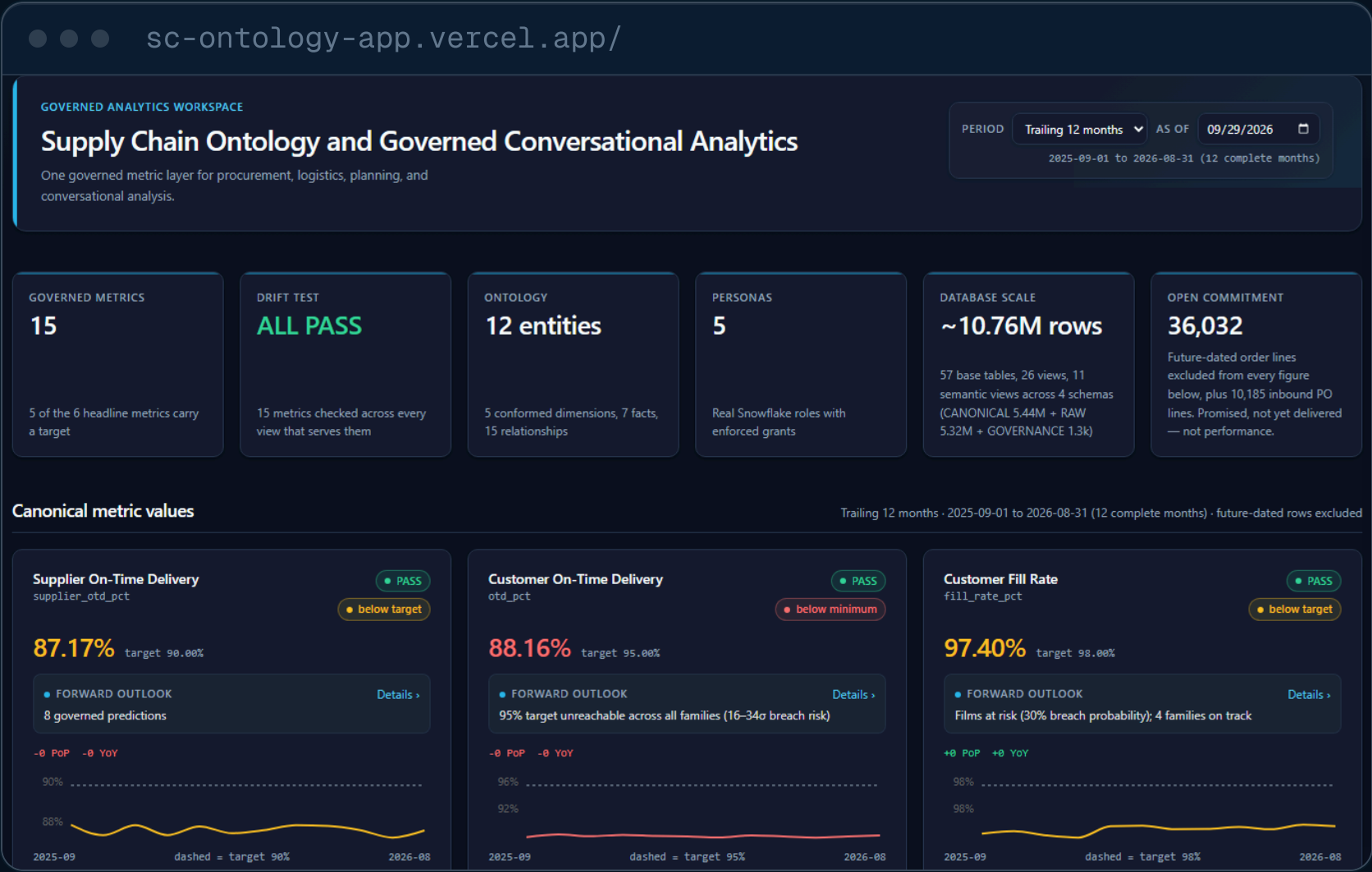
Definition conflicts surface. 79 contracts encode the non-governed definition of on-time.

\$815K
claimable penalties

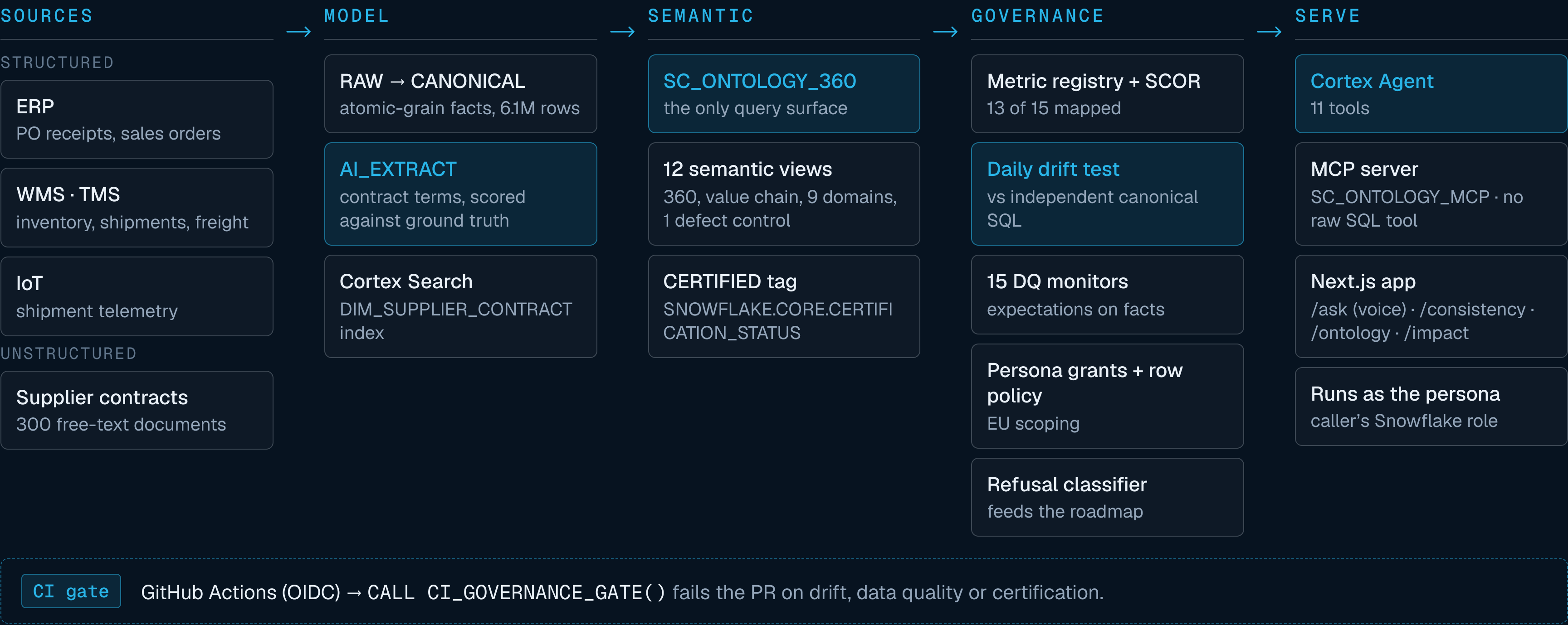
\$179K
hidden by the legacy metric

FEATURE 05 · OPERATIONS

Built for operations, *not just reports.*



Everything queries one surface: *the ontology*.



Fourteen layers, *one skill each*.

LAYER	SQL MODULE	COCO SKILL
Sources and facts	00a-00d, 11	sql-author
Semantic views, agent	00e, 07, 10	agent-studio
Roles, row policy, grants	00a, 00f	data-governance
Drift test, schedule, alert	00f, 04, 06	data-quality · snowflake-tasks · alert
Forecast, breach prediction	07b, 08	machine-learning
Contract extraction	16	document-intelligence · cortex-ai-function-studio
SCOR industry alignment	18	sql-author
Certification, DQ, trust badge	19	certify-object · data-quality
MCP server for other agents	20	sql-author
Refusals → ontology roadmap	21	cortex-ai-function-studio
CI governance gate	22, workflow	ci-cd
Weekly breach digest	23	snowflake-tasks · notification
Impact of changing a view	–	lineage
App, voice, OWL export	app/ , app.yml	snowflake-apps

HOW MODULES PLUG TOGETHER

Three contracts. Nothing else is shared.

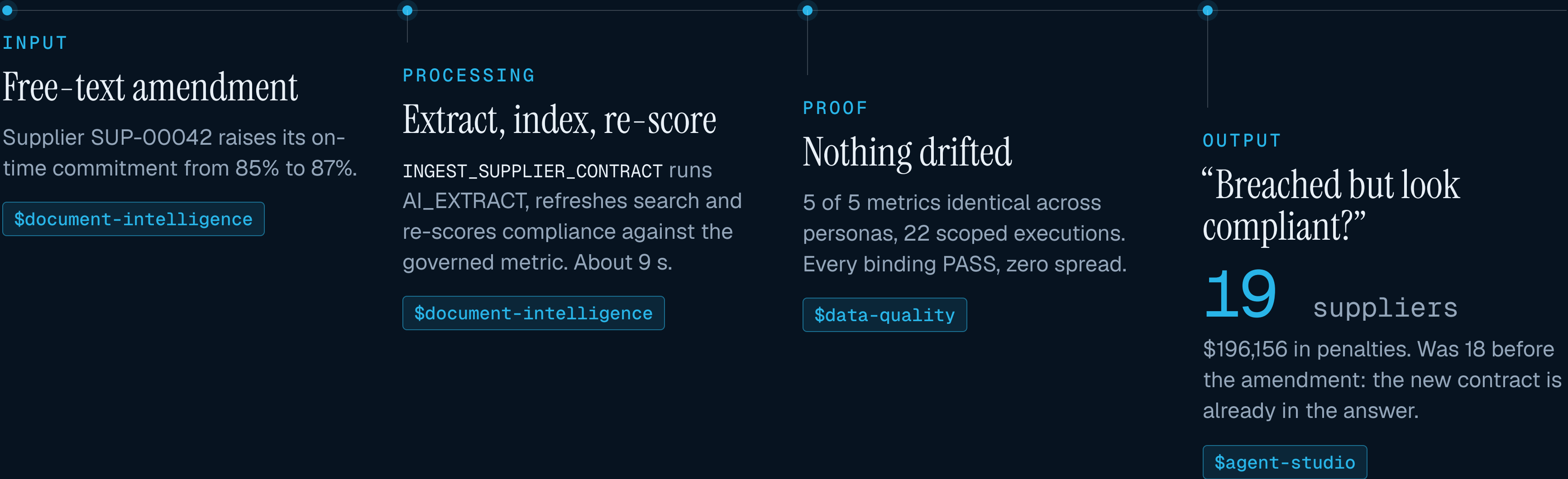
01 Atomic CANONICAL facts
the only source of truth

02 Metric registry
METRIC_DEFINITION +
METRIC_BINDING

03 Persona access
PERSONA_VIEW_ACCESS + row access
policy

rebuild.mjs rebuilds all of it,
6.1M rows, in about 4.5 minutes.

A contract amendment lands. *The answer changes in nine seconds.*



\$178,832

in contract penalties the legacy metric hides, across 18 suppliers.

\$815,222

claimable across 67 suppliers

12

suppliers wrongly cleared

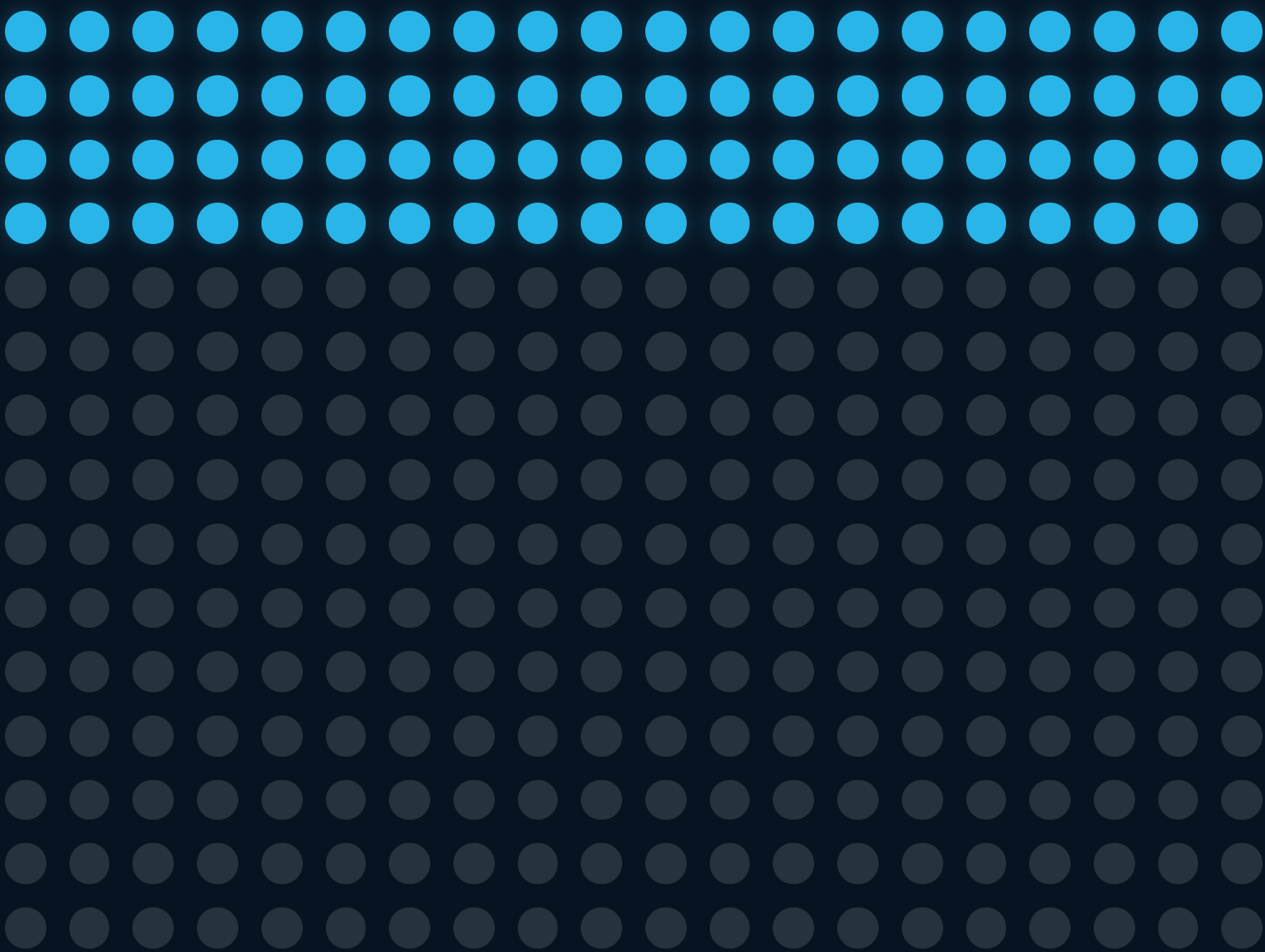
CONSISTENCY	Bindings matching canonical value	15 / 15
CONSISTENCY	Same metric, four personas, own roles	5 / 5
ACCURACY	Conversational eval, 60 questions	93.3%
ACCURACY	Agent vs app vs canonical SQL	4 / 4
ACCURACY	Contract fields extracted correctly	2,700 / 2,700
TRUST	Data-quality monitors passing	15 / 15
GOVERNANCE	EU persona drill-down: EU-only rows, cost masked	18,245 vs 69,667
TIME SAVED	vs 4 h manual reconciliation (assumed)	3,421 _x
TIME	Governed answer, then repeat	4.2 _s → 1.7 _s

Source GOVERNANCE.V_IMPACT_SCORECARD · live at /impact. Data is synthetic; targets are illustrative.

79 of 300 contracts *write the flaw into the agreement.*

They define on-time as a monthly average. The enforceable number and the governed number differ by construction.

A dashboard cannot renegotiate a contract. The ontology can flag every one.



● 79 monthly-average clause ● 221 other

A new metric is *one row*.

A new source is one canonical fact. IoT and contracts were both added that way, without touching the base semantic layer.

- Finance close
- Clinical ops
- Retail

Beyond the demo: the same pattern applies anywhere two teams compute “the same” metric.

NO-CODE PICKUP

Drift test, registry page, consistency runner and /ask all see a new metric. No code change.

THE ROADMAP WRITES ITSELF

Refused questions are classified; unregistered-metric demand is ranked. Forecast totals and schedule adherence lead.

PORTABLE ONTOLOGY

OWL/SKOS export as Turtle or JSON-LD. SCOR codes as exactMatch and closeMatch.

NEW DOCUMENT TYPES

Quality certificates, rate confirmations, customs declarations. Extract, score, route low certainty to review.

EVERY AGENT, SAME NUMBER

MCP exposes governed metrics, executing as the caller. EU logistics 0.8959, global 0.8855.

GOVERNANCE IN THE BUILD

A PR that drifts a binding or certifies the defect view fails CI. Proven red and green.

WHY THIS IS BEST

Proven on *all three* judging focuses.

REAL-WORLD RELEVANCE

Models the chain in the brief and prices the divergence in decisions.

12 suppliers

\$178,832 in penalties the legacy metric hides

TECHNICAL EXECUTION

Definitions are proven, not asserted, and a broken control stays deployed.

15 / 15

zero spread · 252 of 252 tests passing

SOLUTION COMPLETENESS

Structured and unstructured data, scored chat, MCP, a CI gate, OWL export.

93.3%

60-question eval · 4 of 4 parity · 2,700 fields

One definition. Every team.
The same answer, and a
test that proves it tomorrow.

LIVE DEMO

[sc-ontology-
app.vercel.app](https://sc-ontology-app.vercel.app)

TEAM

3M-ONTOLOGIST · Mahantesh,
Devansh, Sathyam

NOTE

The broken definition is left deployed so you can
watch the test catch it.